KPIS_INDITEX_ONETEAM

CENTRALIZED DATA ANALYTICS PLATFORM FOR INDITEX OPERATIONS

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Final Project – Data Analytics MBA. ThePower
Tools used: Power BI, Power Query, Python, Pandas, Excel, DAX

Executive Summary

The objective of this project was to design and implement a **centralized Business Intelligence solution** capable of consolidating operational data related to logistics activities performed for the client **INDITEX**.

Before the implementation of this system, operational data was distributed across multiple teams and departments, each managing their own Excel-based KPI tracking systems. This created significant challenges in terms of data visibility, operational monitoring, and reporting efficiency.

The solution developed in this project integrates all operational KPI files into a unified **Power BI analytical environment**, allowing stakeholders to monitor operations across different teams, countries, and operational areas through interactive dashboards.

The system is based on a scalable architecture that automatically integrates new operational data each year, allowing continuous growth of the dataset.

Using **Power Query for ETL processes, Python for exploratory data analysis, and DAX for advanced calculations**, the project delivers a robust analytical platform that supports operational decision-making and performance monitoring.

1. Business Context

INDITEX is one of the largest fashion retailers in the world, operating multiple brands including:

- Zara
- Zara Home
- Lefties
- Tempe
- Other regional operations

The logistics operations supporting this client involve multiple internal teams located in different countries and responsible for different operational areas such as freight forwarding, ocean transport, delivery tracking, and operational coordination.

Each operational team generates its own **KPI reports**, generally using Excel-based trackers.

These reports monitor operational metrics such as:

- container volumes
- shipment releases
- invoicing
- pre-alerts
- delivery performance
- operational alerts

However, these reports were not connected to a centralized data platform.

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2. Problem Statement

The main problem identified was **data fragmentation across operational teams**.

Within the organization, several teams were simultaneously working on the INDITEX account:

- Control Tower Netherlands
- Control Tower Spain
- Forwarding Spain
- Ocean freight operations
- ECO delivery monitoring

Each of these teams maintained their own reporting structures and KPI files.

This situation generated several operational challenges:

- **Lack of Global Visibility.** Management could not easily obtain a consolidated view of all operations related to the INDITEX account.
- **Manual Reporting.** Monthly reports require manual consolidation of multiple Excel files.
- **Inconsistent Data Structures.** Different teams used different formats and column definitions.
- **Limited Historical Analysis.** Comparing operational trends across multiple years requires extensive manual work.
- **Limited Data Exploration.** Excel-based reports limit the ability to explore operational data dynamically.

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4. Project Objectives

The main objective of the project was to build a **centralized analytical system** capable of integrating all operational KPI data.

Specific objectives included:

1. Integrate multiple operational Excel files into a single analytical environment.
2. Standardize data structures across teams.
3. Build a scalable architecture capable of incorporating future data.
4. Enable interactive analysis through dashboards.

5. Improve visibility of operational performance.
6. Support monthly operational meetings with automated reporting.

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5. Data Architecture

The analytical solution was built around a **folder-based data architecture**, where operational teams maintain their KPI files in structured directories.

The main Power BI report file is:

KPIS_INDITEX_ONETEAM.pbix

This file connects to several folders containing the operational data sources.

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6. Data Sources

The dataset integrates operational information from multiple teams.

Control Tower – Netherlands (CT NL)

Files included:

KPIs_2025_CTNL_OCEAN
KPIs_2026_CTNL_OCEAN

These files track ocean container volumes managed by the Netherlands Control Tower.

Control Tower – Spain (CT ES)

Files included:

KPIs_2025_ES
KPIs_2026_ES

These files track operational coordination activities handled by the Spanish Control Tower (LEFTIES, ZARA HOME, TEMPE & MARRUECOS pages.

Forwarding – Spain (FW ES)

This operational area contains three main datasets.

- **CESIONES**

KPIs CESIONES 2025
KPIs CESIONES 2026

These files track operational handovers between teams.

- **FACTURACIÓN**

KPIs FACTURACIÓN 2025
KPIs FACTURACIÓN 2026

These files track invoicing operations related to shipments.

- **PREALERTAS**

KPIs Prealertas 2025
KPIs Prealertas 2026

These files monitor operational alerts related to shipments.

OCEAN

Ocean freight operations include:

ECO_Delivery_Master_Report_2025
ITX – MAERSK OCEAN DASHBOARD

These datasets monitor ocean freight shipments volumes and green fuel used on these transports.

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7. Data Integration Strategy

A key objective of the project was to create a **living dataset** capable of expanding over time.

The integration strategy was based on:

- standardized Excel templates

- folder-based data ingestion
- Power Query transformations
- yearly data append processes

Each year, new KPI files are added to the corresponding folders using the same naming convention.

Example:

KPIs_2027_CTNL_OCEAN

Power BI automatically integrates the new files during dataset refresh.

This design ensures **long-term scalability** of the analytical platform.

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8. Data Preparation (Power Query)

The ETL process was implemented using **Power Query**, which allowed the transformation and cleaning of the data before loading it into the analytical model.

Key transformation steps included:

1. **Column Standardization.** Column names were standardized across all datasets.

2. **Data Cleaning.** Several cleaning operations were performed:

- removing duplicated rows
- fixing inconsistent text values
- handling null values
- correcting date formats

3. **Data Enrichment.** Additional columns were created to improve the analytical capabilities of the dataset.

   Examples include:

   - brand classification

- operational region
- shipment category

EXAMPLE:

CTs MASTER DATA

```
let
    // 1️⃣ Carga todos los archivos de la carpeta (sin depender de Sample File)
    Source = Folder.Files("C:\Users\JMA705\OneDrive - Maersk Group\Inditex Control Tower - Documentos\Control Tower - NETHERLANDS (CT NL)\OCEAN FLOW\KPIS - Ocean\POWERBI_UPDATED"),
    VisibleFiles = Table.SelectRows(Source, each [Attributes]?[Hidden]? <> true),

    // 2️⃣ Aplica la función de transformación existente a cada archivo
    TransformFiles = Table.AddColumn(VisibleFiles, "Transform File", each #"Transform File"([Content])),

    // 3️⃣ Mantén solo las columnas necesarias
    KeepColumns = Table.SelectColumns(TransformFiles, {"Name", "Transform File"}),

    // 4️⃣ Expande la tabla transformada de cada archivo
    ExpandedData = Table.ExpandTableColumn(KeepColumns, "Transform File", Table.ColumnNames(#"Transform File"(VisibleFiles{0}[Content]))),

    // 5️⃣ Cambia tipos según tu definición original
    ChangedType = Table.TransformColumnTypes(ExpandedData,{
        {"Name", type text}, {"VESSEL", type text}, {"TERMINAL", type text}, {"ATA", type datetime}, {"ETA", type datetime}, {"HAULER", type text},
        {"MASTER BL", type any}, {"CARRIER", type text}, {"PUERTO ORIGEN", type any}, {"CONTAINER", type text}, {"WAREHOUSE", type text},
        {"DELIVERY AVAILABLE SINCE", type datetime}, {"TRUCK SHOW UP", type datetime}, {"TIMESLOT", type datetime}, {"DC start date", type datetime},
        {"TRANSPORT MODE", type text}, {"CLEANING", type text}, {"EXTRA STOP", Int64.Type}, {"WEEK", Int64.Type}, {"MONTH", Int64.Type},
        {"HORA DE LLEGADA CONT", type datetime}, {"ATA A OFRECIMIENTO", type number}, {"ATA A OFRECIMIENTO 2", type number}, {"AVAILABILITY  -48H", type text},
        {"DEL. AVAILABLE A SLOT", type number}, {"DEL. AVAILABLE A SLOT 2", type number}, {"TIMESLOT MENOS DE 24H ", type text}, {"ATA A TIMESLOT", type number},
        {"ATA A TIMESLOT 2", type number}, {"PUNTUALIDAD (H)", type number}, {"PUNTUALIDAD 2", type text}, {"PUNTUALIDAD 3", type number},
        {"ATA TO UNLOADING", type number}, {"ATA TO UNLOADING 2", type number}, {"DELIVERED IN -48H", type text}, {"COMMENTS FOR LATE DELIVERY AND FOR DELAYS", type text},
        {"HOUR", type datetime}, {"NIGHT", type text}, {"CONTAINER4", type text}, {"VENDOR", type text}, {"COST", Int64.Type}, {"CLEANING5", type number},
        {"E.S. COST", type number}, {"EXTRA STOP TOTAL", type number}, {"WEEKDAY", Int64.Type}, {"DELIVERED AT", type text},
        {"VENDOR+BASECOST", type text}, {"TOTAL COST", type number}, {"ADMIN FEE", type number}, {"WAITING TIMES", type any}, {"WAITING TIMES (HOURS)", type any},
        {"TOTAL SELL", type number}, {"MANAGEMENT FEE", Int64.Type}, {"ALTIA", Int64.Type}, {"ESTIMATION COMPLETED ON", type text}
    }),
   #"Removed Columns" = Table.RemoveColumns(ChangedType,{"ESTIMATION COMPLETED ON", "WAITING TIMES (HOURS)", "Name", "COMMENTS FOR LATE DELIVERY AND FOR DELAYS", "HOUR", "WEEK2", "CONTAINER4", "VENDOR"}),

#"Added FLUJO" = Table.AddColumn(#"Removed Columns", "FLUJO", each "NETHERLANDS", type text),

CleanedData = Table.ReplaceErrorValues(#"Added FLUJO",{
    {"CLEANING5", 0}, {"ADMIN FEE", 0}, {"TOTAL SELL", 0}, {"COST", 0}, {"E.S. COST", 0}, {"TOTAL COST", 0}
}),
    #"Appended Query" = Table.Combine({CleanedData, #"CT ES: data append"}),
    #"Filtered Rows" = Table.SelectRows(#"Appended Query", each true),
    #"Removed Columns1" = Table.RemoveColumns(#"Filtered Rows",{"MONTH", "WEEK"})
in
    #"Removed Columns1"
```

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9. Exploratory Data Analysis using Python

Before building the final dashboards, an exploratory data analysis phase was conducted using **Python and Pandas**.

This analysis helped validate the integrity of the data and detect potential inconsistencies.

Typical operations performed included:

```
import pandas as pd

df = pd.read_excel("KPIs_2025_CTNL_OCEAN.xlsx")

df.describe()

df.groupby("Month")["Containers"].sum()
```

The exploratory analysis allowed the identification of:

- data distributions
- potential outliers
- missing values
- operational patterns

This step ensured that the dataset was ready for visualization.

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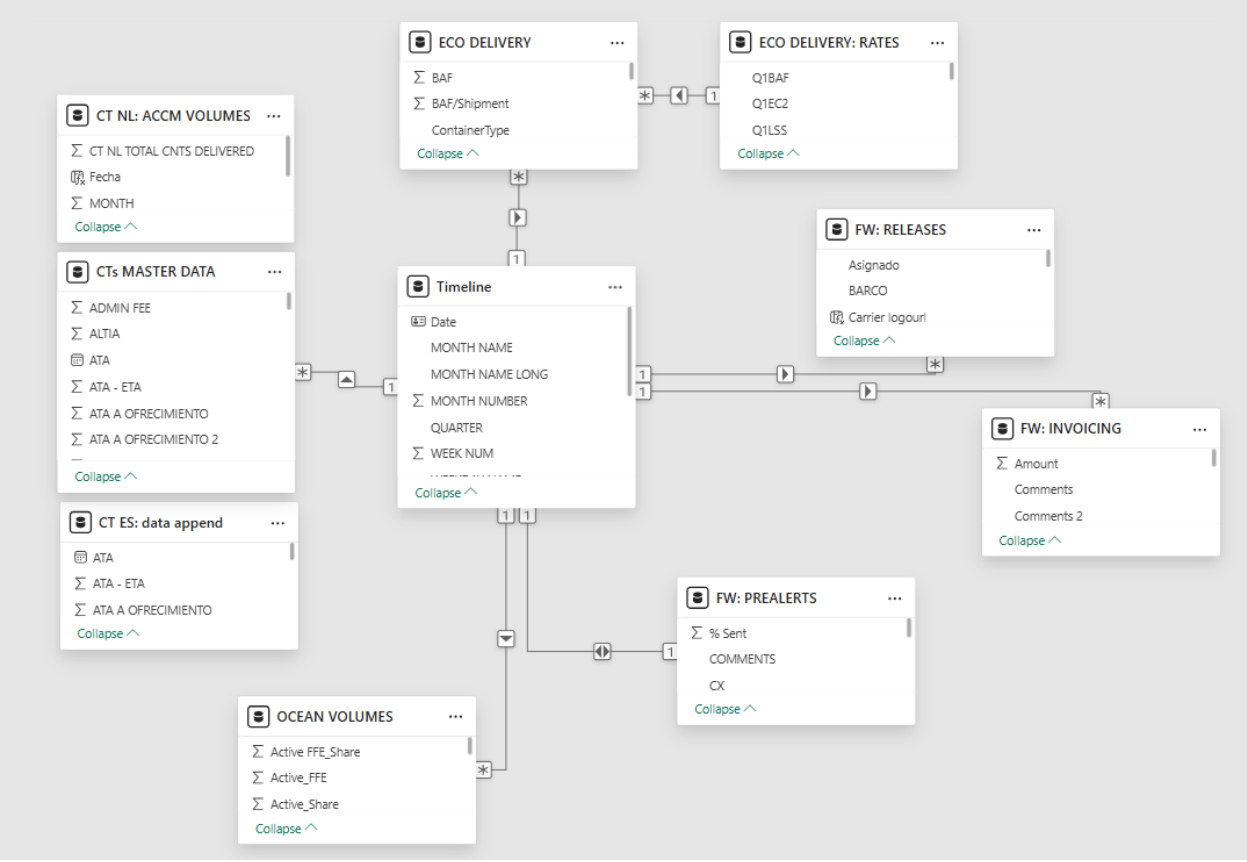
10. Data Model Design

The analytical model was designed using a **Star Schema architecture**.

This design improves performance and simplifies analytical queries.



Central Table: Timeline

The **Timeline table** acts as the central dimension of the model.

Fields include:

Date
Year
Month Number
Month Name
Quarter

All operational tables are connected to this timeline.

Fact Tables

The model includes several fact tables representing operational data:

CT NL ACCM VOLUMES (Chose to preserve this in case in the future we prefer to have this splitted from ES).
CT ES DATA APPEND (same for this one)
FW RELEASES
FW INVOICING
FW PREALERTS
ECO DELIVERY

These tables store transactional operational data.

Dimension Tables

Supporting dimension tables include:

CTs MASTER DATA
ECO DELIVERY RATES

OCEAN VOLUMES
Timeline

These tables provide attributes used for filtering and grouping.

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11. Data Model Relationships

The data model is structured around the timeline dimension.

Example simplified structure:

Timeline
|
|------ OCEAN VOLUMES
|
|------ FW RELEASES
|
|------ FW INVOICING
|------ FW PREALERTS
|
|------ CT NL ACCM VOLUMES
|
|------ CTs MASTER DATA
|
|------ ECO DELIVERY

This structure enables time-based filtering across all operational datasets.

```
Timeline =

VAR MinDate =
    MINX(
        UNION(
            SELECTCOLUMNS('CTs MASTER DATA',"Date",'CTs MASTER DATA'[TRUCK SHOW UP]),
            SELECTCOLUMNS('ECO DELIVERY',"Date",'ECO DELIVERY'[Date Equivalent]),
            SELECTCOLUMNS('OCEAN VOLUMES',"Date",'OCEAN VOLUMES'[Ocean_Arrival_Date]),
            SELECTCOLUMNS('FW: RELEASES',"Date",'FW: RELEASES'[Date Added]),
            SELECTCOLUMNS('FW: INVOICING',"Date",'FW: INVOICING'[Invoice Issue Date Clean]),
            SELECTCOLUMNS('FW: PREALERTS',"Date",'FW: PREALERTS'[DATE TIME DEADLINE])
        ),
        [Date]
    )

VAR MaxDate =
    MAXX(
        UNION(
            SELECTCOLUMNS('CTs MASTER DATA',"Date",'CTs MASTER DATA'[TRUCK SHOW UP]),
            SELECTCOLUMNS('ECO DELIVERY',"Date",'ECO DELIVERY'[Date Equivalent]),
            SELECTCOLUMNS('OCEAN VOLUMES',"Date",'OCEAN VOLUMES'[Ocean_Arrival_Date]),
            SELECTCOLUMNS('FW: RELEASES',"Date",'FW: RELEASES'[Date Added]),
            SELECTCOLUMNS('FW: INVOICING',"Date",'FW: INVOICING'[Invoice Issue Date Clean]),
            SELECTCOLUMNS('FW: PREALERTS',"Date",'FW: PREALERTS'[DATE TIME DEADLINE])
        ),
        [Date]
    )

RETURN

ADDCOLUMNS(
    CALENDAR(MinDate, MaxDate),
    "YEAR", YEAR([Date]),
    "MONTH NUMBER", MONTH([Date]),
    "MONTH NAME", FORMAT([Date],"mmm"),
    "MONTH NAME LONG", FORMAT([Date],"mmmm"),
    "QUARTER", "Q" & FORMAT([Date],"Q"),
    "WEEK NUM", WEEKNUM([Date],2),
    "WEEKDAY NUMBER", WEEKDAY([Date],2),
    "WEEKDAY NAME", FORMAT([Date],"dddd"),
    "YEAR-MONTH", FORMAT([Date],"YYYY-MM"),
```

12. DAX Measures

Several DAX measures were created to support advanced analysis.

Some examples:

Total Containers

Total Containers =
SUM('OCEAN VOLUMES'[Containers])

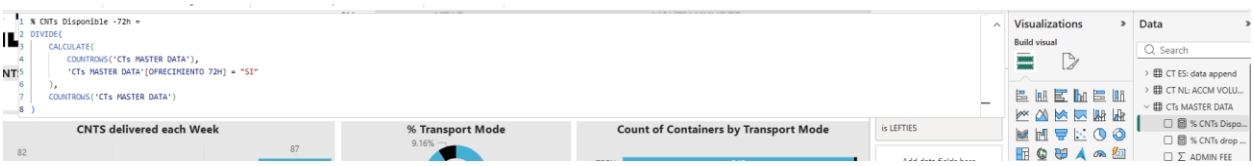
Unique Containers

Unique Containers =
DISTINCTCOUNT('OCEAN VOLUMES'[Container_ID])

Monthly Volume

Monthly Containers =
CALCULATE(
SUM('OCEAN VOLUMES'[Containers]),
ALLEXCEPT(Timeline, Timeline[Month])
)

Or



These measures allow dynamic analysis of operational volumes.

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13. Dashboard Design

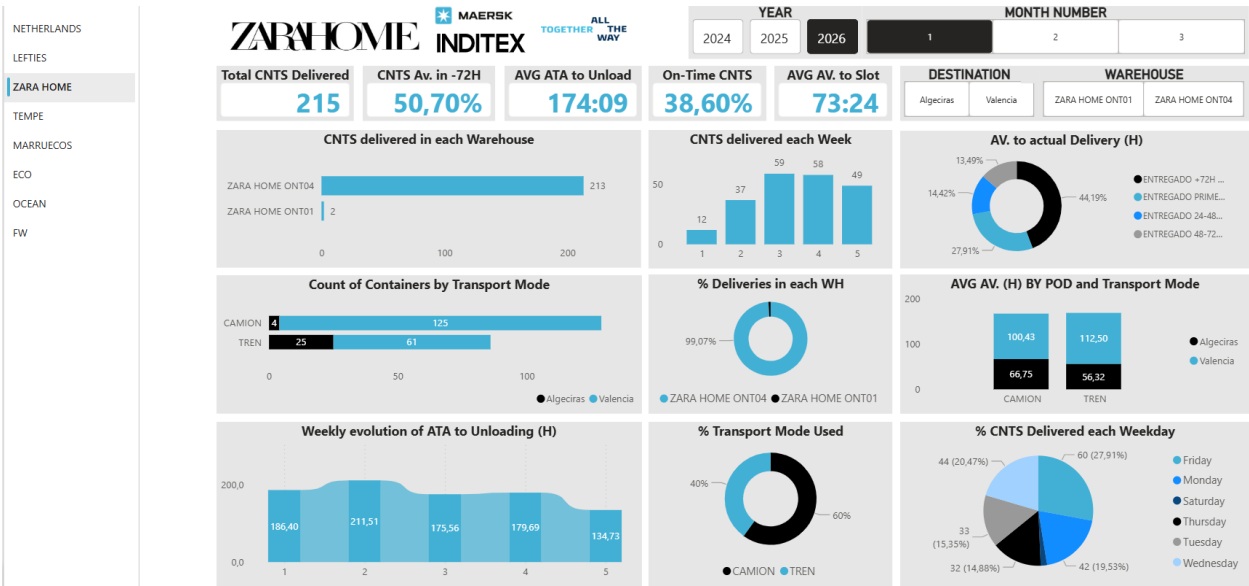
The Power BI report contains several dashboard pages designed for different operational areas.

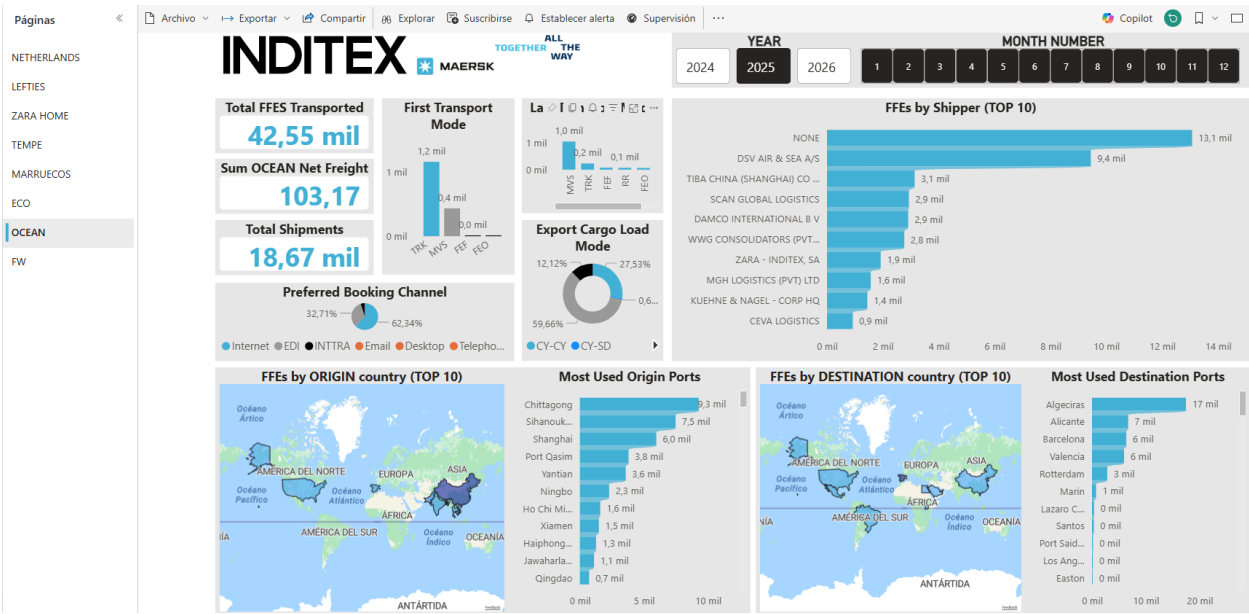
Main pages include:

- Netherlands
- Lefties
- Zara Home
- Tempe
- Marruecos
- ECO
- Ocean
- Forwarding

Each dashboard page includes:

- KPI indicators
- volume trends
- operational breakdowns
- interactive slicers





14. Operational Use Case

The dashboard is used during **monthly operational meetings** with internal stakeholders.

The dashboards allow teams to review:

- shipment volumes

- operational performance
- monthly trends
- team activity levels

Users can filter the data by:

- month
- brand
- operational port
- warehouse
- ETC

This allows flexible exploration of operational data.

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15. Key Benefits

The implementation of this analytical platform delivered several benefits.

a) **Centralized Data.** All operational KPIs are now consolidated in a single system.
b) **Reduced Reporting Effort.** Manual reporting work has been significantly reduced.
c) **Improved Decision Making.** Management can now monitor operational performance in real time.
d) **Historical Analysis.** The system allows analysis of trends across multiple years.

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16. Future Improvements

Several improvements could further enhance the analytical platform.

Potential future developments include:

- automation of Excel report generation
- integration with operational databases
- predictive analytics for shipment volumes

- machine learning models for delay prediction

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17. Conclusion

The **KPIS_INDITEX_ONETEAM project** successfully demonstrates how modern data analytics tools can transform fragmented operational data into a centralized business intelligence platform.

Through the integration of Power BI, Power Query, Python, and DAX, the project delivers a scalable and efficient analytical environment that significantly improves operational visibility.

The platform provides stakeholders with powerful tools to monitor logistics performance and support data-driven decision making.